

RELATIVE QUANTUM GRAVITY

Localized Gravity and the Swampland

Relative quantum gravity: A gravity theory which violate the Swampland Conjectures, but satisfy them when defined as a quantum theory coupled to a higher-dimensional gravity theory.

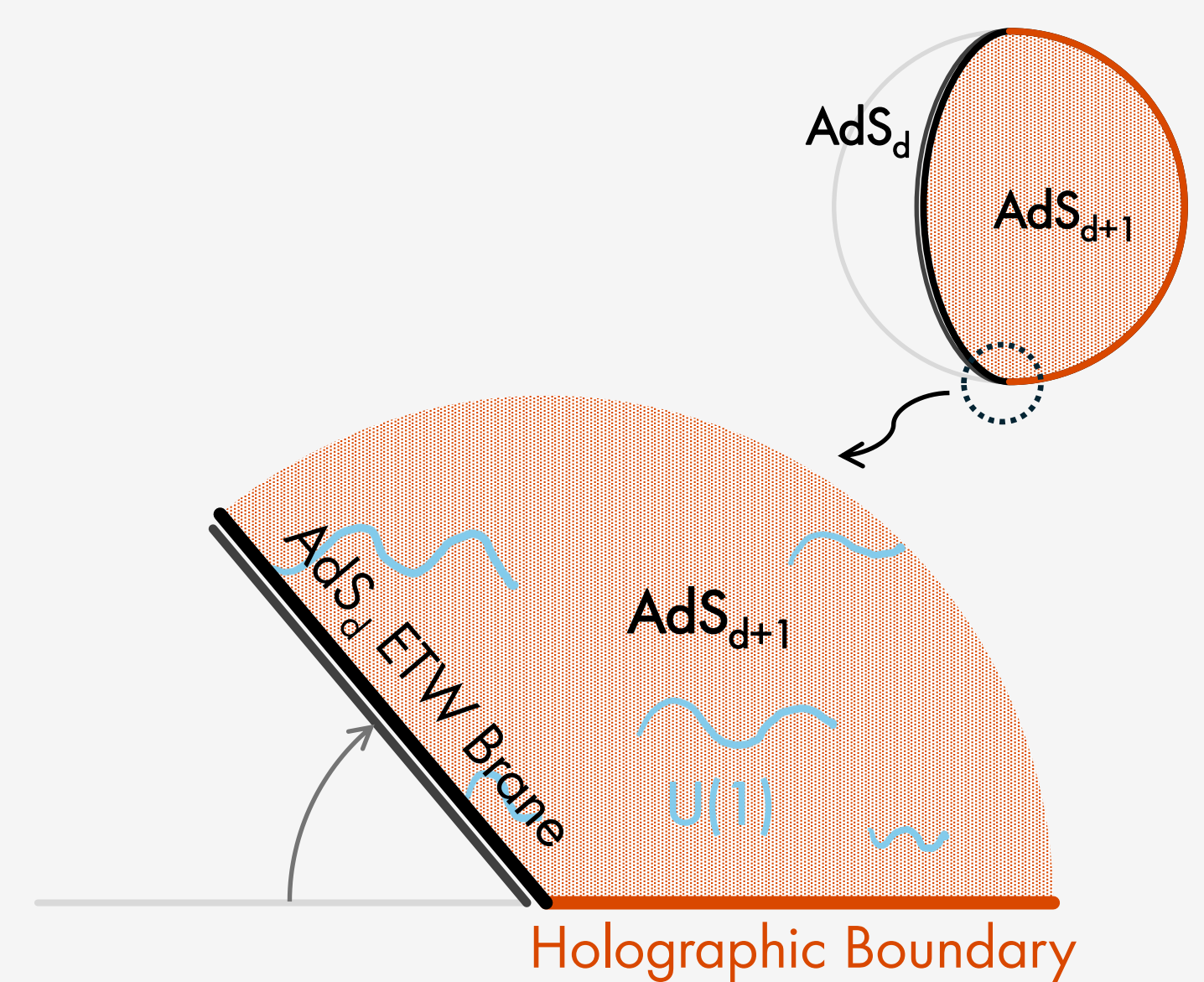
LOCALIZED GRAVITY IN KARCH-RANDALL

There are many ways to localize gravity in a submanifold of spacetime. The most prominent one is the Karch-Randall setup. It consists in an AdS_d End of the World (ETW) brane with tension, living on AdS_{d+1} , which allows a massive mode of the $(d+1)$ graviton to localize on the d -dimensional brane.

However, gauge fields are not localized, and it is seen as a global symmetry from the QG theory living on the brane. This violates the Swampland **No Global Symmetry** conjecture!!

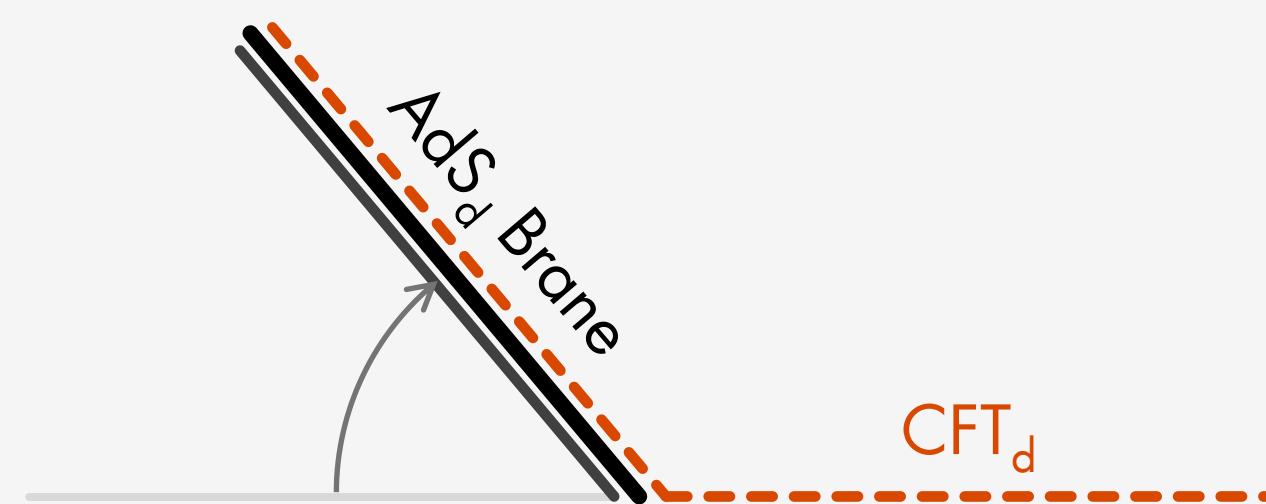
Moreover, we can localize gravity on a de Sitter slice in Randall-Sundrum setups, so we may ask ourselves why and how these constructions are unstable, as "predicted" by the **de Sitter Conjecture**.

Maybe these Swampland constraints can be violated on relative theories of Quantum Gravity as long as they are, well, relative.



THE "INTERMEDIATE" PICTURE

Double Holography: Because of holography, we have an "intermediate" description, where the above setup is equivalent to gravity on the AdS_d brane coupled to a CFT_d with a UV cutoff, which is the dual of the bulk AdS_{d+1} gravity.



The action on the brane is:

$$S_d = \frac{1}{16\pi G_d} \int_B d^d x \sqrt{-h} \left[R + \frac{2}{\ell_d^2} + \frac{\ell^2}{(d-2)(d-4)} \left(R_{\mu\nu} R^{\mu\nu} - \frac{dR^2}{4(d-1)} \right) + \dots \right] + I_{CFT}$$

The dynamical gravity is given by the region integrated out, and ℓ controls the series expansion. This coefficient is also the UV cutoff of the CFT. The central charge of this CFT_d reflects the number of degrees of freedom (or species), and is related to the parameter ℓ as

$$\ell \sim (c_d G_d)^{\frac{1}{d-2}}$$

In these setups, for $d=3$, there have been extensive research on black holes, in particular the ones called "quantum black holes". The radius of these BHs have a size proportional, again, to ℓ .

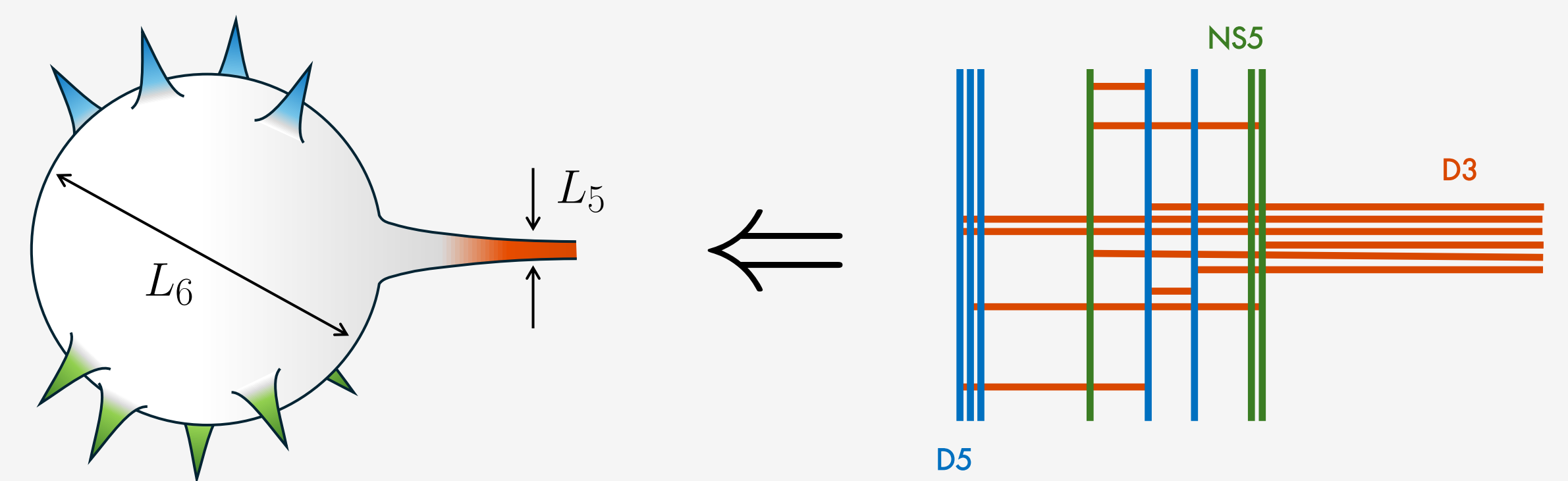
The Species Scale: It is a proposed universal cutoff for any EFT that is given by:

- The number of light degrees of freedom
- The radius of the smallest black hole
- The parameter that controls the expansion in higher curvature terms

We see that this "intermediate" picture indeed provides a quantitative characterization of the Species Scale, in particular in a strong coupling context with a large number of degrees of freedom, and allows to connect the different definitions in the literature.

The Emergence Proposal: If we introduce a gauge symmetry in the bulk, it has to be global in the CFT side. But in the "intermediate" picture, as the CFT has a cutoff (due to the region that has been integrated out), we get a gauge symmetry. So we see that we get the gauge kinetic terms integrating out modes, that is, lowering the cutoff (in this case, the Species Scale). The setup then provides an explicit quantitative realization of the Emergence Proposal. We can check that when we send $\ell \rightarrow 0$, the gauge coupling goes to infinity, and we recover a global symmetry.

RELATIVE QUANTUM GRAVITY IN STRING THEORY



The gravitational dual of D3s ending on D5- and NS5-branes provides a String Theory example of Relative Quantum Gravity. We have a bag, $AdS_4 \times X^6$, with an $AdS_5 \times S^5$ spike given by the backreaction of the D3s. For $L_6 \gg L_5$, we don't see the AdS_5 spike at low energies, we have gravity localized on the bag.

Notice that here we don't have scale separation. The **AdS Distance Conjecture** is then naively satisfied at our Relative QG theory. But all these ideas can in principle be applied to setups with scale separation.

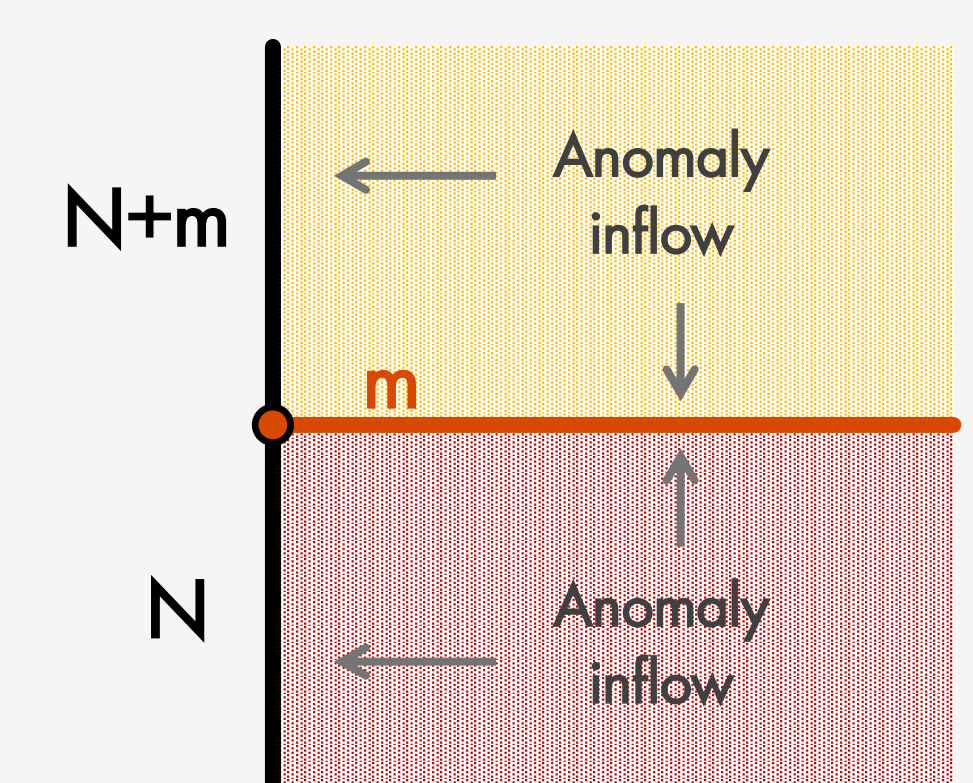
The **Completeness Conjecture** states that in a consistent quantum gravity theory, for every gauge field there must exist all possible electric and magnetic sources compatible with Dirac quantization condition. If we neglect the semi-infinite AdS_5 spike, we will find that this conjecture (as well as the **Weak Gravity Conjecture**) is violated on the following way:

The only states charged under the worldvolume gauge symmetry of a D5-brane stack are fundamental strings with endpoints on the D5-branes, but these are in the adjoint representation, hence neutral under the center of mass $U(1)$. So we don't have states in the bag charged under this gauge symmetry. To have charged states we need strings with only one endpoint on the D5-brane stack, so we need the non-compact direction provided by the AdS_5 spike.

Then, we see that although our Relative Quantum Gravity violates some Swampland constraints, when completed with the extra dimension, everything becomes consistent.

RELATIVE QFT

These constructions are related to the interesting setups of relative QFTs, where anomalous QFTs are made consistent by regarding them as boundaries of higher dimensional theories with an anomaly in the bulk. Both anomalies cancel due to anomaly inflow and the whole system is consistent.



We can induce anomalies in the bag for these String Theory solutions (also in the KR setup by hand). We can check that then the **Cobordism Conjecture** is not satisfied on the Relative QG, as we can forbid 't Hooft anomaly matching, having global symmetries with different anomalies.

Then, to connect two such d -dimensional Relative QG and satisfy the Cobordism Conjecture, we have to join at the $(d-1)$ -dimensional interface a d -dimensional relative defect that extends onto the bulk.

This nicely relates with the intuition that to satisfy the Swampland constraints we need to take into account the extra dimension, or that an anomalous theory is made consistent if we couple it to a bulk with an anomaly inflow.



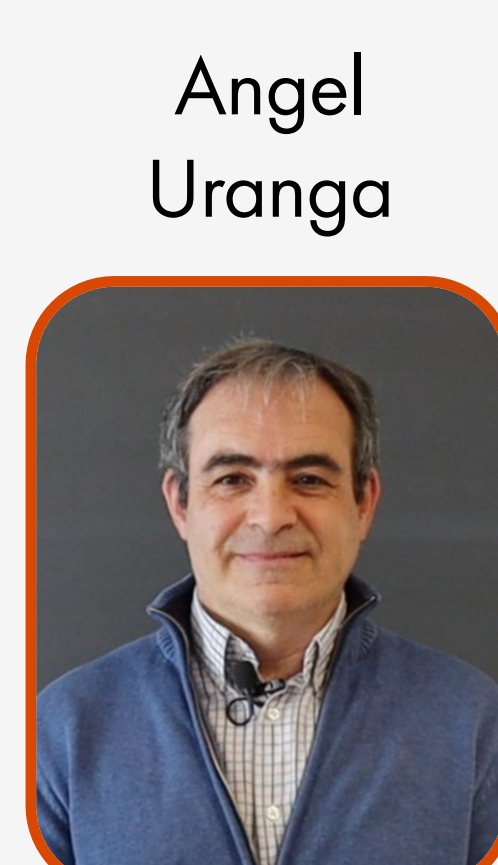
Edoardo
Anastasi



Roberta
Angius



Jesús
Huertas



Angel
Uranga



Chuying
Wang



SCAN ME!

This poster is based in
2501.03310



Instituto de
Física
Teórica
UAM-CSIC